

Artificial intelligence and software design

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High Performance
Real Time **Lab**

**I TUOI
FUTURI MEDICI
STANNO USANDO
CHATGPT PER
SUPERARE GLI
ESAMI DI MEDICINA**



AI– A brief recap

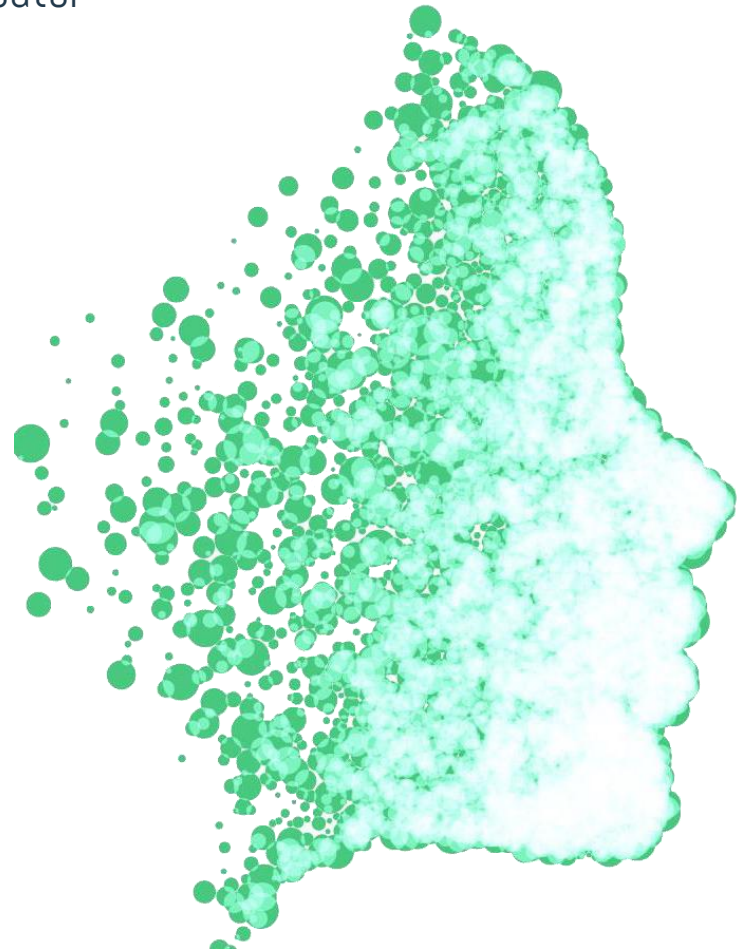
Instead of “giving orders” to the computer about what to do (traditional informatics, based on algorithms)...

- › Reproduce (and improve!) a brain inside a computer
- › Train it, and it learns
- › Not anymore deterministic



DISCLAIMER

AI is taught later in MS, here I will explain some basic concepts that might be helpful to you in your career as dev





(My) open questions

Where to apply AI?

- › Single component vs. e2e

Use AI for code dev

- › You teach me than more I can teach you
- › Vibe coding? Or co-pilots? (Implications on the job model)

Can we structure the SW development process to enable scalability?

- › Vibe coding?
- › Enforce design patterns? What about SOLID and CLEAN?
- › Use AI in design? And testing?



Pros and cons

Pros (countless)

- › Increase productivity
- › Automate not only code dev, but all processes
- › ...

Cons

- › Not deterministic
- › Require datasets
- › Replaces workforce (that cannot work without AI)
- › ...



The DORA report from Google *

"In 2025, the central question for technology leaders is no longer if they should adopt AI, but how to realize its value."

Not anymore a novel disruptive technology, it's a reality

› ..and it's here to stay!

I am not saying you can't do research on AI..

› ..what I am saying is that, once the hype effect is gone, we need to approach AI in a **systematic and structured** manner



DORA's main outcome

"AI's primal role in SW development is the one of an amplifier"

- › It magnifies the strengths and dysfunctions of your organization/team

The **returns on AI investment** comes from the underlying organizational system

- › The *quality* of the internal platform
- › The clarity of *workflows*
- › The alignment of teams

The risk

- › Creating localized pockets of productivity that are often lost to downstream chaos



DORA report - Key aspects

AI is pervasive

- › 90% of respondents use AI
- › 80% declare it increases productivity
- › 30% do not trust results from AI

You shall not simply adopt tools

- › It's about practices, and workflows
- › (It's about having a plan, and not having the AI drive you)

AI improves SW delivery throughput and pace

- › But it also increases SW instability
- › Teams and single devs are using it effectively for speed
- › Orgs and their processes cannot keep the pace



DORA report - Key aspects (cont'd)

Value stream management (VSM) as an effective practice (“force multiplier”)

- › Continuous monitoring and improving the workflow from idea/design to to customer
- › Suitable for Agile methodologies, with fast Time-To-Market (hence, AI)
- › Translates team productivity into **measurable** improvements

Identified seven type of team profiles

- › We'll see this later

Monitoring is key to avoid disasters

- › Need to identify metrics and KPIs referring to the process, not to the technology!



AI for coding



..you teach me!

Question: How do you use AI?





..you teach me!

Question: How do you use AI?

I use it for

- › Prototyping and scaffolding (for instance, I am not good at all at UX programming..)
- › Debugging: error detection & prediction
- › Support for punctual problems
 - *"How do I configure an Apache server to reject all GET requests?"*
 - *"Can you scaffold a Visitor pattern with 4 classes and 30 visitors, with all interfaces?"*

Don't confuse the two things!

Increase one's productivity **VS.** *"quick & dirty" I-don't-know-what-I'm-doing approach*

(The latter is ok for non-experts in the field, who need to do prototyping)



Copilot* vs Vibe coding

Besides prototyping... how can you effectively build code using AI assistants?

- › Answer: this is part of your “freedom” so, it’s up to you
- › ..if you still are in the “frame” of **good design, good processes** (, ethics) and company policies

Business models for LLMs – from assistants to AI Agents

- › ChatGpt – Generic but mainly B2C
- › MS Copilot – B2C
- › Claude – B2B

Paid vs. free

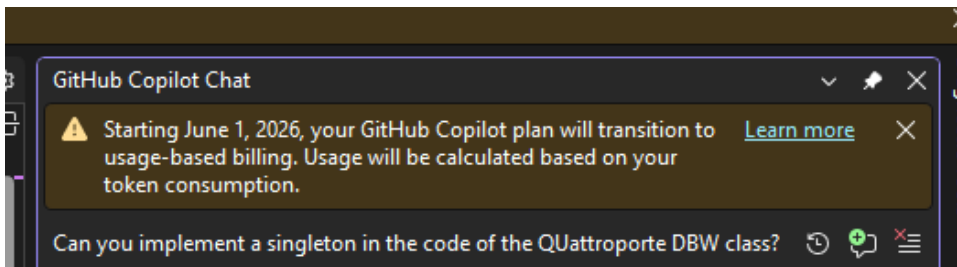
- › Basically, better models
- › And training vs. no training - Your company shall not share code, nor expertise, nor data

**Disclaimer: this is not about Microsoft Copilot 😊*



Only one hint for you...

Let the AI be a tool/friend,
not your replacement



- Mi dica cosa sa fare che non possa
fare con ChatGPT





How?

There is still something that AI needs... and will always need

A context

- › Domain information "context"
- › Goals
- › Ground (dataset)



AI agents

Beyond “simple” helpers, they can accomplish complex tasks

- › Up to a full coding + testing + delivery pipeline

Key features

- › Still, they need you to define the context and goal
 - the agent decides the sequence of actions needed to reach it
- › Easy integration with external software
 - Databases Svc APIs, business applications, other agents)
- › Planning & Memory
 - They self-manage the process

Best for

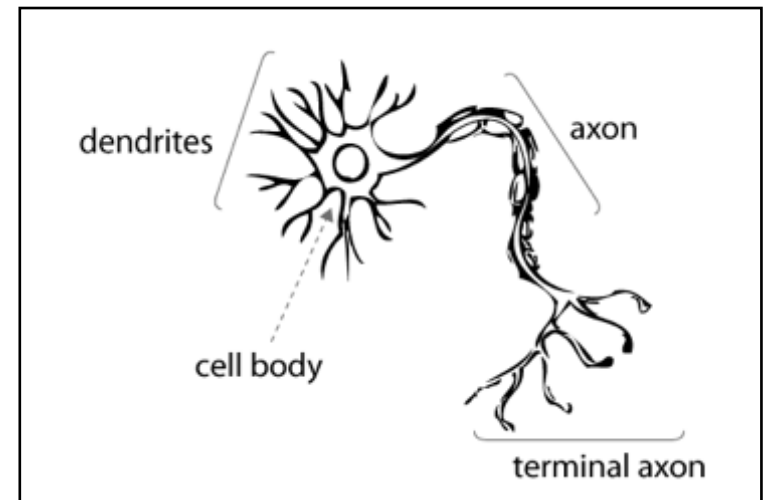
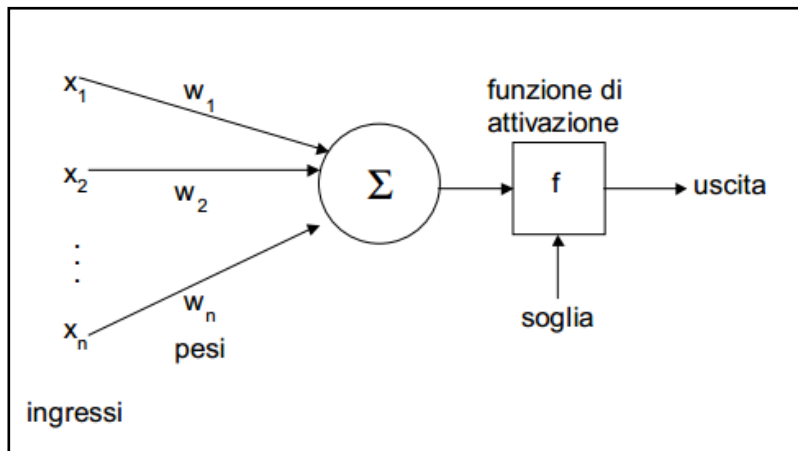
- › Customer Support & troubleshooting large systems
- › Business Operations and systems integration by automating data workflows



Designing AI systems

Convolutional Neural networks

- › Bio-inspired
- › Based on neurons arranged in layers
 - And sub-layers
- › Convolutional neural network
 - Perform Convolutions

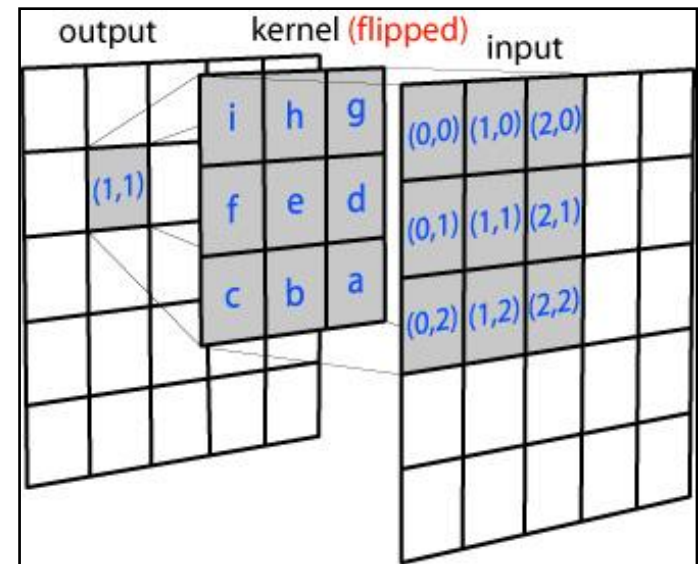




Convolution

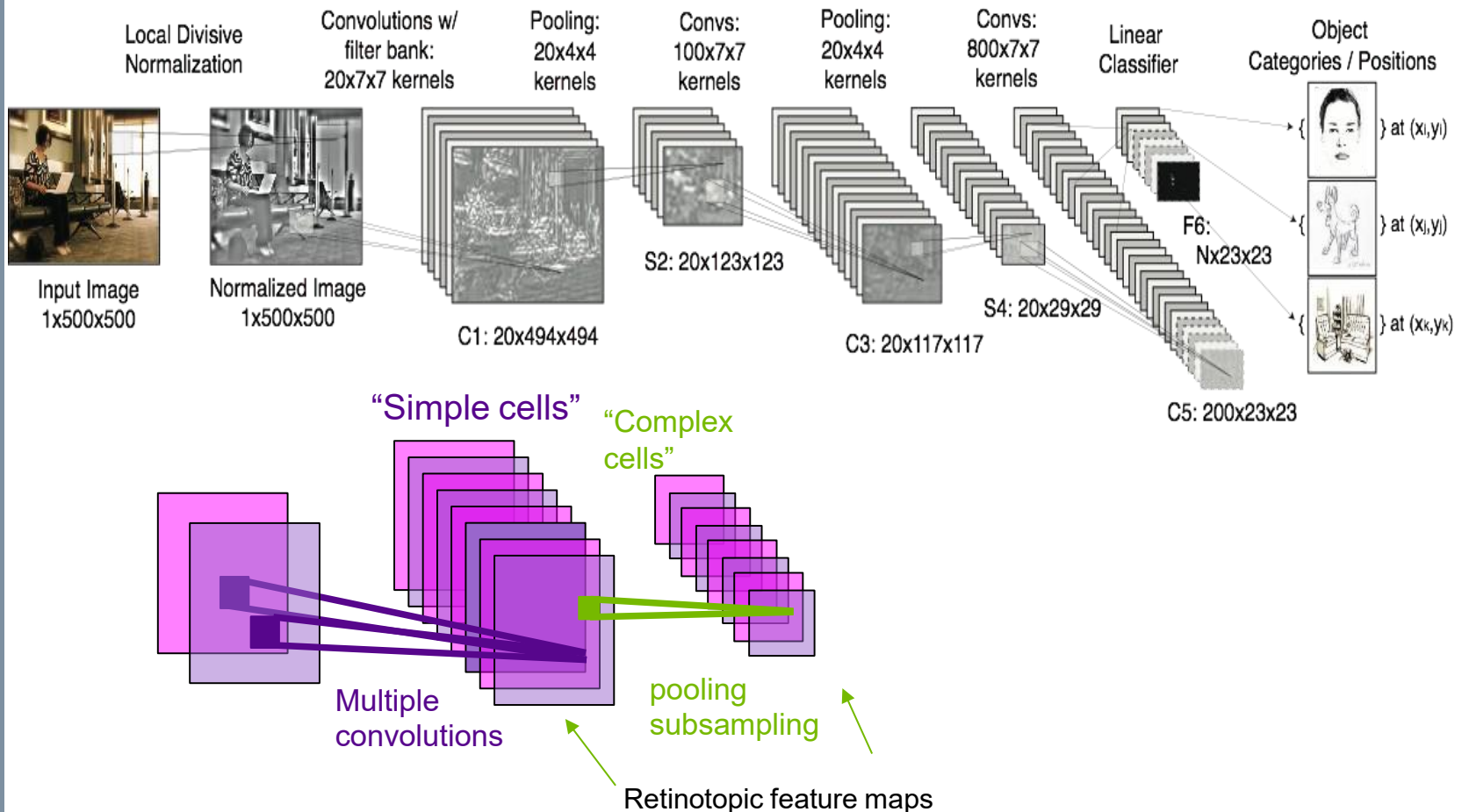
- › Computation-intensive
- › Suitable for implementation in hardware
- › In computer vision, blurring

$$\begin{aligned}(f * g)(t) &\stackrel{\text{def}}{=} \int_{-\infty}^{\infty} f(\tau)g(t - \tau) d\tau \\ &= \int_{-\infty}^{\infty} f(t - \tau)g(\tau) d\tau.\end{aligned}$$



The convolutional net model

> (Multistage Hubel-Wiesel system)





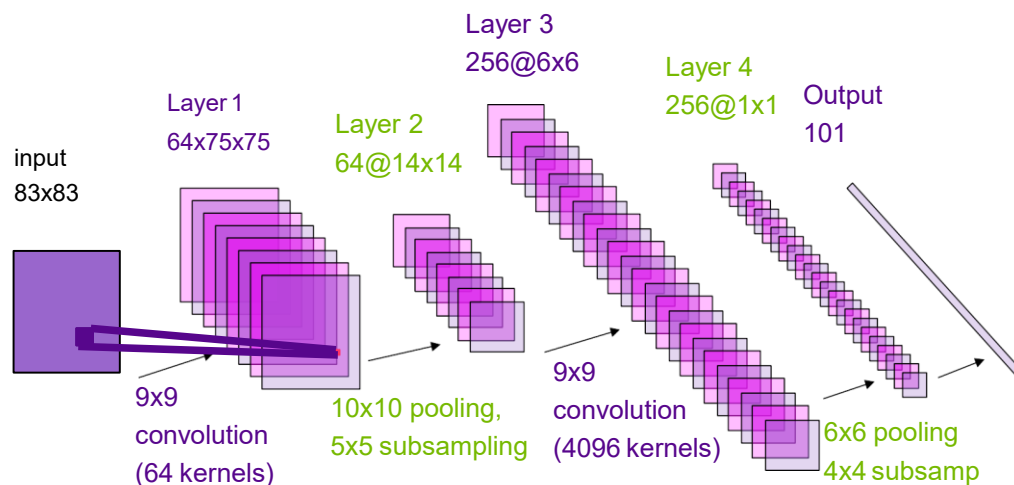
Challenge 1 – building the network

Network topology

- › How many layers and sublayers?
- › How big they are?
- › How are they connected?

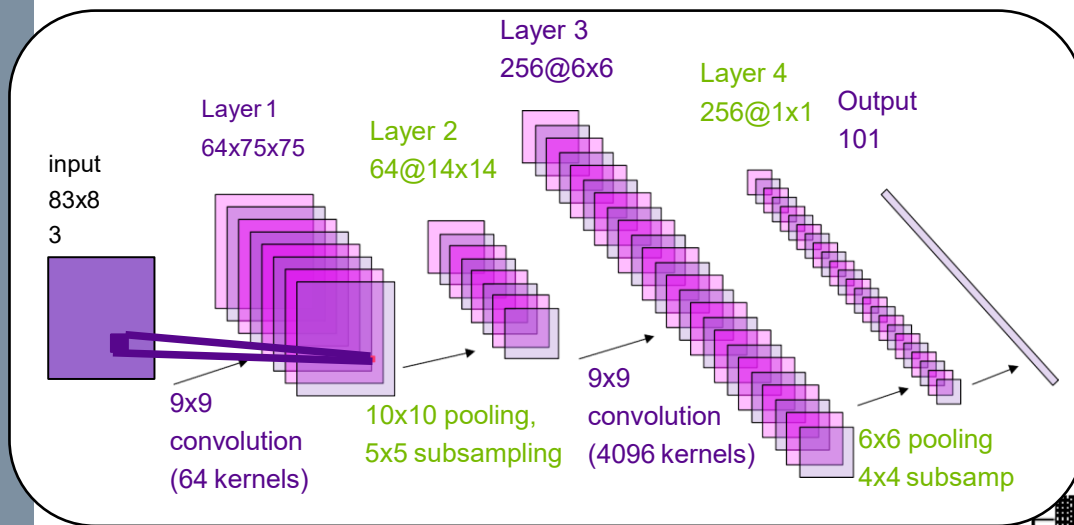
Neuron type

- › CNN
- › int/float datatypes
- › How to perform pooling?

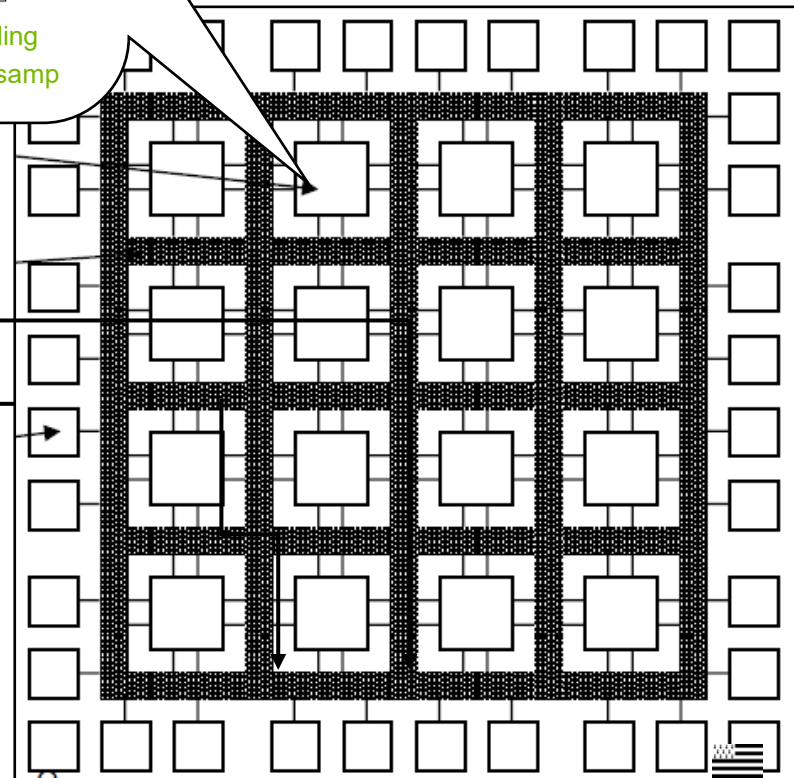
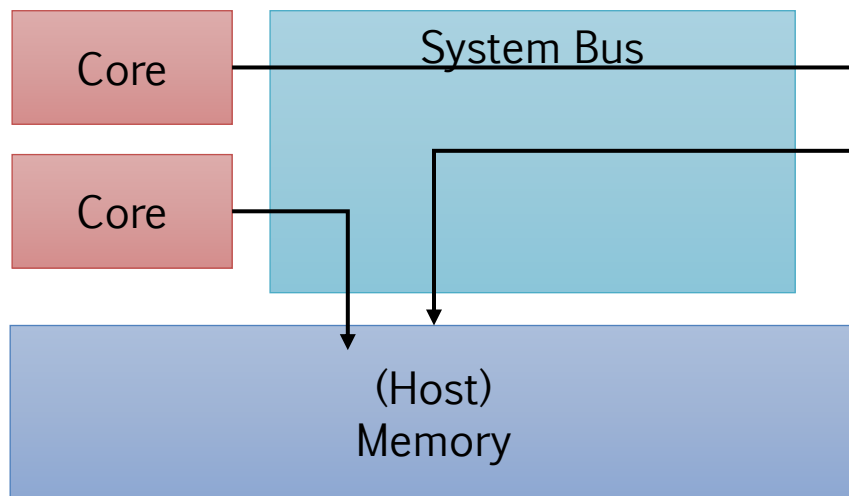




Challenge 2 – which computer?



> Here, an embedded board, for inference

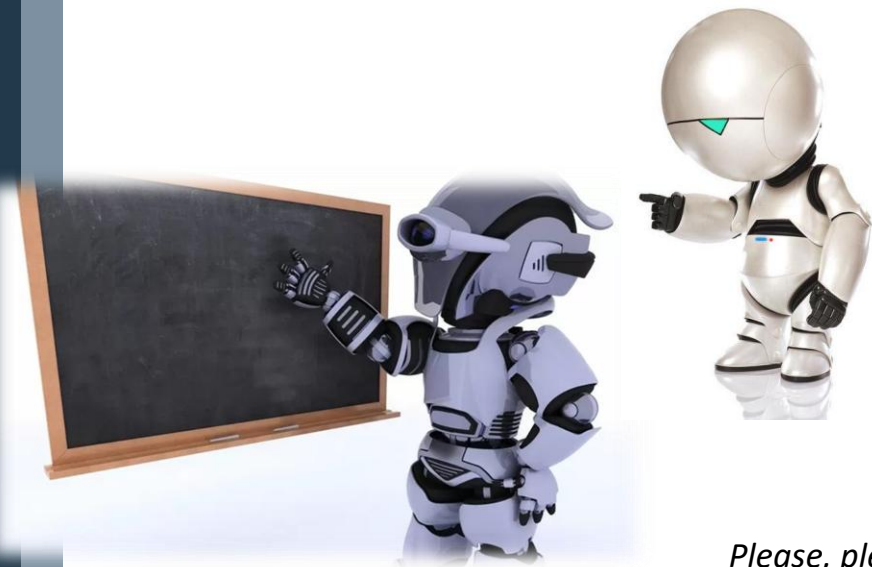




Machines go to school...

...machines learn!

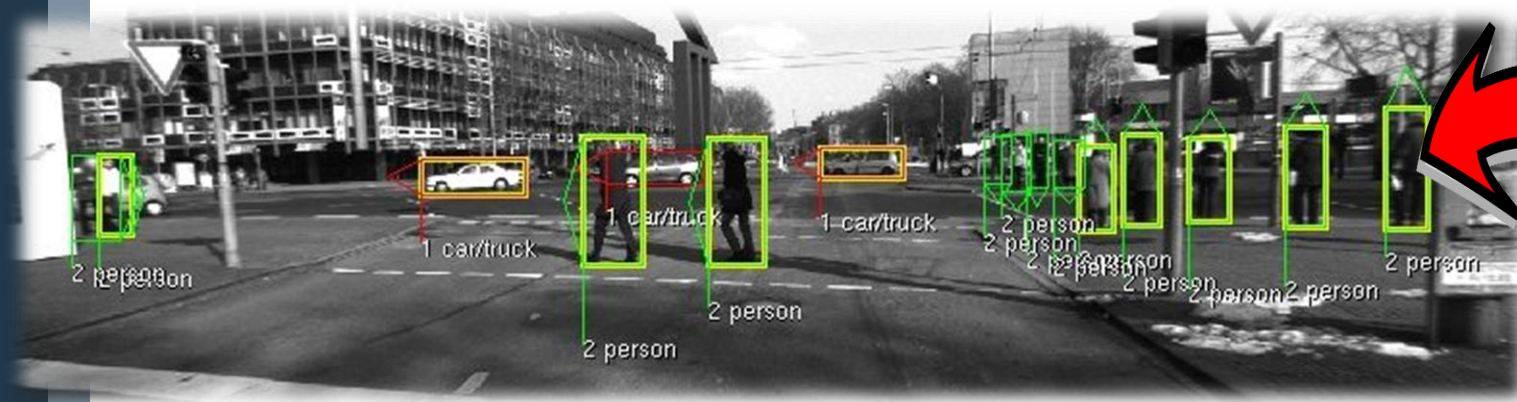
- ✓ Near-human performance
- ✓ Surely, better than other SW artifacts
- ✓ There is no other (efficient way)



*Please, please, **please**, leave Skynet out of the discussion, ok?*



Challenge 3 - Training sets





We need real data!

- › Why do you think Google does self-driving cars?
- › Why do you think big cloud players want our data?
- › Collect, store and manage data in a safe and GDPR-compliant manner





How to design a dataset?

Two approaches, not exclusive

- › Model centric (Focus on the network model) vs. data centric (create/improve the dataset)

What you need, first

- › A comprehensive view of how your model works
- › The computational requirements of your use case (are we running on servers? But then we need to transmit the data!)
- › SLA / timing requirements (is the user interaction async or sync?)



Key concepts to dataset design

Data quality

- › Data must be relevant to the context, consistent (no duplicates..), and compliant to privacy, law, ethics

Data quantity

- › Enough, but too much can lead to biases

Data coverage

- › Shall cover all the requirements of your system
- › (e.g., we plan to deploy a system that predicts traffic flows in Italy, but not in China)

Optimization

- › For the accuracy, for the target computing platform..



Example: data coverage in Llama 3

	Pre Training	Supervised FineTuning	Preference FineTuning
General Knowledge	50%	52.66%	82%
Math and Reasoning	25%	21.19%	6%
Coding	17%	15%	7%
Multilingual	8%	3%	5%
Exam like	X	8%	x
Long Context	X	0.1	X



Data lifecycle: an iterative process



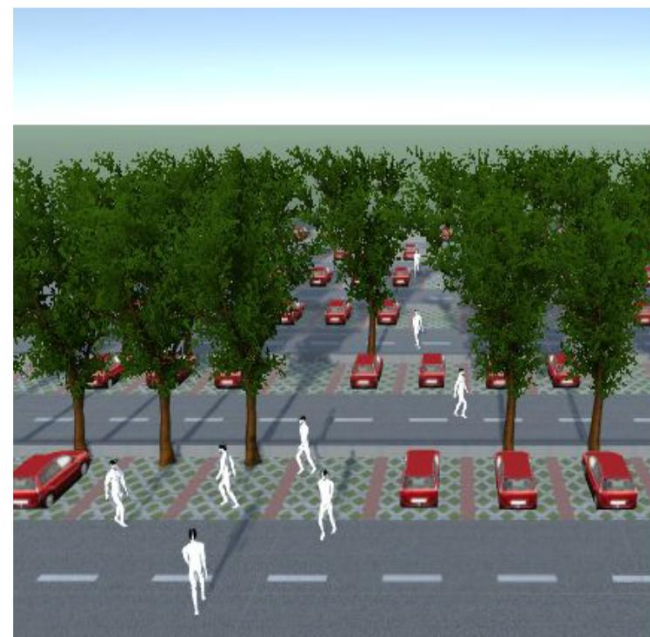
Improving your data

Data Augmentation

- › Creating variations of existing data to improve robustness and reduce bias.
- › “Paolo is drinking water” – “Mario is drinking water”

Data Synthesis

- › Generating new data, by randomly modifying existing data
- › Example: random pedestrian generation in for autonomous parking scenario (Bachelor Thesis, y2018)
- › Disclaimer. Random is usually pseudo-random! You insert this in the model





Human-in-the-Loop (HITL)

Def: "A model requiring human interaction"

- › Typically, during a modeling & simulation process
- › Used in testing to detect corner cases, and to test the robustness to unpredictable cases

In Machine learning

- › During the building/training phases of a model
- › Humans help selecting the most critical data, and corner cases

Pros/cons

- › Performs better than random techniques (unlike our previous example ☹)
- › Requires time and effort



Managing uncertainty



It's statistics, baby!

MATHEMATICS

IS

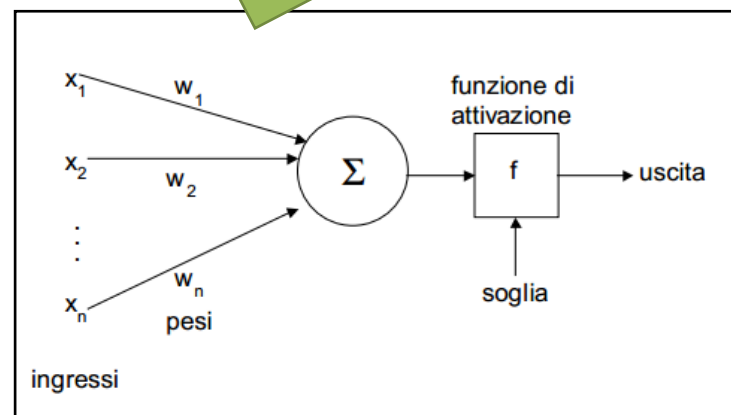
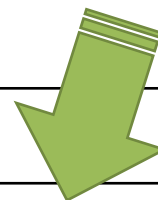
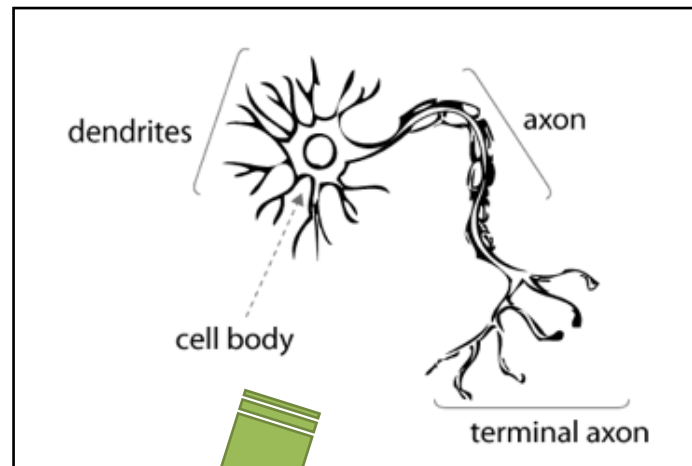
GEOMETRY

PROBABILITY

ALGEBRA

STATISTICS

CALCULUS

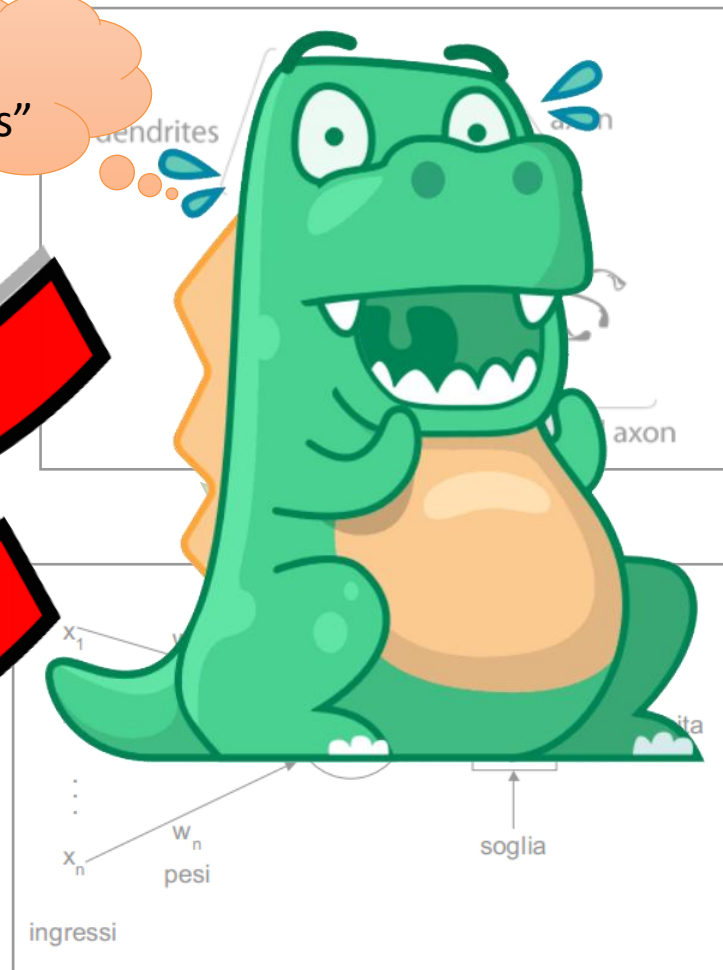




It's statistics, baby!

MATHEMATICS
IS
GEOMETRY
PROBABILITY
ALGEBRA
STATISTICS
CALCULUS

"It works
99.5% times"





"I conti della serva"

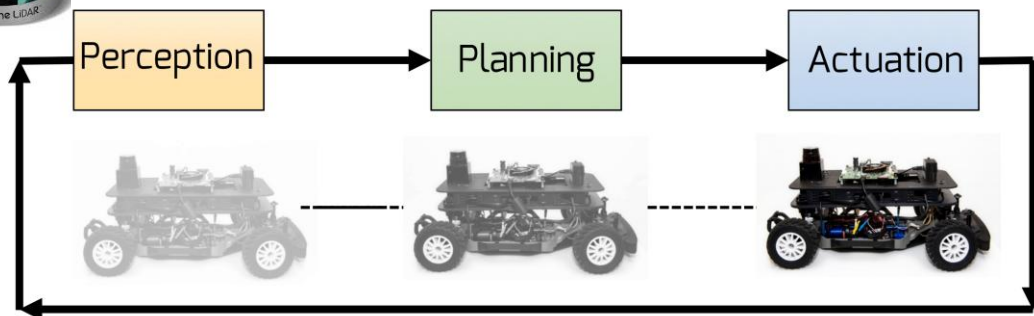
Fails once every 200 times

=> Fails every 2 seconds

"It works
99.5% times"



10ms/100Hz





AI components monitoring and explainability

AI is inherently non-analyzable

- › ..or, at least, it's very difficult to understand "what's in one's mind"

Still, we might need to monitor while running

- › AI must be "explainable" in the sense that I need some indicators to measure its performance
- › Trivial solution, analyzing how subset of the model (layers, or neurons) behave during training
- › Employing redundancy at system level (e.g., voting systems)
- › Monitor the Operational Design Domain, i.e., if the context stays within given parameters
 - "When it rains, my detection accuracy drops from 98% to 85%"





AI for design scalable systems



Areas

- › Requirements collection
- › System design & integration
- › Coding
- › Testing
- › Team management



Requirements collection

Trivial: you can use AI as tools

- › Read.ai for meeting minutes
- › To create powerpoint presentations and docs

(But this is simple! Can we do more?)

AI could directly interact with customer

- › ...can it? 😊

...but...

- › AI can predict what customers want
- › (With statistically meaning accuracy)

So, it can support you with useful information about your customer needs

- › Based on previous customers
- › “This guy is a tech guy, so he will provide you more tech information and you will need to elicit system-level information”

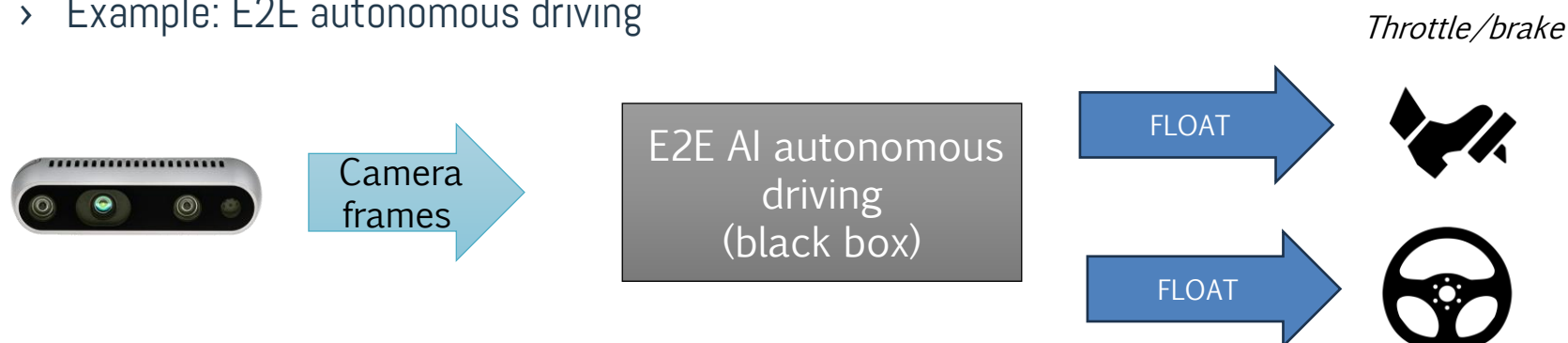




AI as software foundation

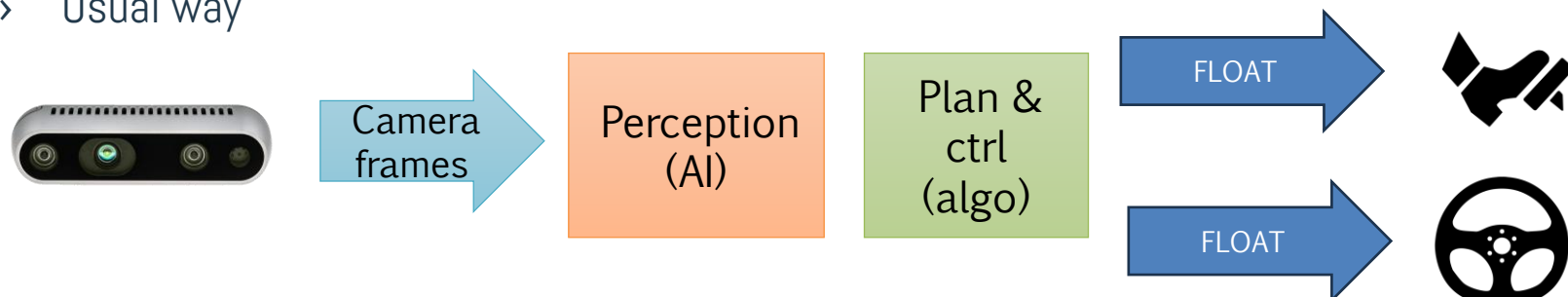
End-to-end approaches

- › Your system is 100% based on AI. There are no algorithmic/deterministic components
- › Also called “Behavioral cloning”
- › Example: E2E autonomous driving



Integrating AI components

- › “Usual way”





Testing

Straightforward from requirements analysis, we can generate tests automatically

- › Several tools exists
- › E.g., <https://www.getxray.app/>, <https://www.testrail.com/>
- › Luckily, if a test fails due to wrong AI decision, we are forced to “look into it”, and refine

Remember: using AI means propagating mistakes and uncertainties from your development phases

- › If requirements are not good (e.g., they don't cover all use cases) neither will your system
- › So, the requirements collection must be well structured and oriented, e.g., to build a dataset
 - Check out the *Cucumber* language <https://cucumber.io/>
- › “I your gym club, do you have customers of less than 18?”



AI and teams/processes



DORA - Software delivery performance metrics



Throughput

- › #changes / time
- › E.g., commits, LoC, push requests

VS.

Instability

- › #issues / time
- › E.g., “issues”, bugs (which now can be automatically detected” 😊)





Seven team profiles / clusters

1. Foundational challenges (10% overall)

- › Problems in adopting technology
- › Burnout and frictions
- › Software is not stable

2. Legacy bottleneck (11% overall)

- › Poor performance, still can deliver updates
- › Burnout and frictions
- › Software becomes stable after a number of iterations



Seven team profiles / clusters (cont'd)

3. Constrained by process (17% overall)

- › Create low value, but still create it
- › Burnout and frictions
- › Software is stable – it's not about competences, but working environment/processes

4. High impact, low cadence (7% overall)

- › Very good productivity
- › Low frictions
- › The environment is instable, harnessing long-term stability

5. Stable and methodical (15% overall)

- › “Artisans”: excellent productivity, however at low pace
- › Healthy environment
- › Software and environment are stable



Seven team profiles / clusters (cont'd)

6. Pragmatic performance (20% overall)

- › Good software, good pace
- › Average burnout and frictions, but low engagement ("I work because you pay me but no more than this")
- › Software and environment are stable

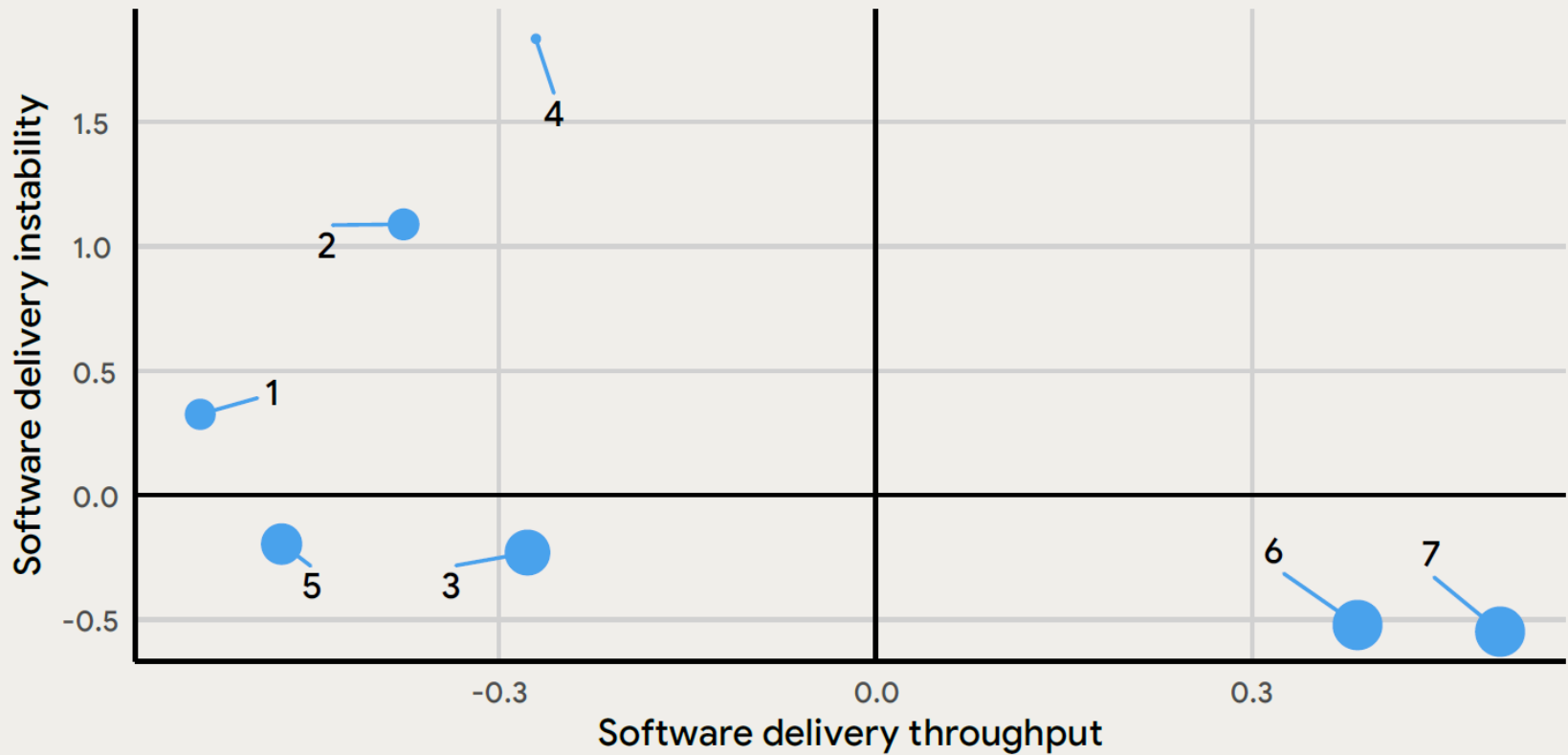
7. Harmonious high-achiever (20% overall)

- › Good software, good pace
- › Low burnout and frictions
- › Software and environment are stable



At a glance

Cluster distribution across software delivery performance factors





References



Course website

- › <http://hipert.unimore.it/people/paolob/pub/ProgSW/index.html>

The DORA report: State of AI-assisted Software Development

- › <https://dora.dev/research/2025/dora-report/>

My contacts

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